Input = output + losses

$$\% \text{ Efficiency } (\eta) = \frac{\text{output power}}{\text{input power}} \times 100$$

$$= \frac{\text{output power}}{\text{output power + losses}} \times 100$$

(iron loss + copper loss)

$$= \frac{\text{input power} - \text{losses}}{\text{input power}} \times 100$$

6.6.2 Why Transformer Rating in KVA?

As seen Cu loss of a transformer depends on current and iron loss depends on voltage. Hence total transformer losses depends on volt ampere (VA) and not on phase angle between voltage and current i.e, it is independent of load power factor. Therefore rating of transformer is in KVA and not in KW.

$$KVA = \frac{KW}{\cos\theta}$$

6.6.3 Why Transformer does not Work on DC Supply?

The transformer works on the principle of mutual induction, for which current in one coil must change uniformly. If DC supply is given, the current will not change due to constant supply and transformer will not work. This may cause the burning of windings due to extra heat generated and may cause permanent damage to the transformer. Thus DC supply should not be connected to the transformers.

6.7 TESTING METHODS OF TRANSFORMER

There are two tests are performed on a transformer to determine the power. They are:

1. Open circuit test
2. Short circuit test

These two tests are used for finding out the power loss occurring in the transformer.

6.7.1 Open Circuit Test on Transformer

The open circuit test on transformer is used to determine core losses in transformer.

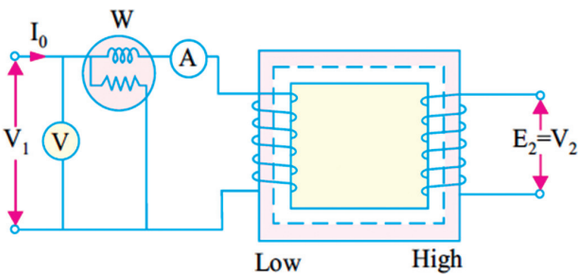


Fig 6.11 *Open Circuit Test*

The connection diagram for open circuit test on transformer is shown in Figure 6.11. A voltmeter, wattmeter, and an ammeter are connected in primary side of the transformer. The voltage at rated frequency is applied to the primary side with the help of a variac (or) variable ratio auto transformer.

The secondary side of the transformer is kept open. Now with the help of variac, applied voltage gets slowly increased until the voltmeter gives reading equal to the rated voltage of the primary side. After reaching at rated primary side voltage, all three instruments reading **(voltmeter, ammeter and wattmeter readings)** are recorded.

The ammeter reading gives no load current (I_o) being secondary is open the transformer draws very less current. Hence copper loss are negligible. As no load current

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